

Colby Frison

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Education

Bachelor of Science in Computer Science

University of Oklahoma, Gallogly College of Engineering

- GPA: 3.76/4.00; President's Honor Roll (Spring 2025)
- Relevant Coursework: Compiler Construction, Data Structures & Algorithms, Software Engineering, Computer Organization, Programming Languages

Expected December 2026

Norman, OK

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript, HTML/CSS, PowerShell

Technologies: Power Automate Desktop, UiPath, React, Next.js, Node.js, Express, Firebase, MongoDB, Tailwind CSS, Git, GitHub, CI/CD

Concepts: Compiler Design, AST Transformations, SSA, RPA, Process Automation, Full-Stack Development, Machine Learning

Work Experience

IT Intern, Automation & AI Development

ConocoPhillips

May 2026 – July 2026

Bartlesville, OK

- Enabled Power Automate adoption by evaluating platform gaps against existing RPA tools and supporting enterprise platform integration
- Migrated existing UiPath workflows into Power Automate Desktop pilot projects to validate feasibility and guide platform decisions
- Documented capability differences across RPA platforms to inform evaluation and adoption planning for the Automation & AI team

Engineering Ambassador

University of Oklahoma, Gallogly College of Engineering

March 2025 – Aug. 2025

Norman, OK

- Served as sole Computer Science ambassador, communicating complex technical concepts to prospective students and families
- Led campus tours of engineering facilities and provided detailed information about engineering majors and departmental opportunities
- Conducted research on peer institutions and high schools to inform college recruitment strategies
- Supported Engineering Days and high school camps through workshop guidance, event operations, and student engagement

Teacher/Teacher's Assistant

Houston Museum of Natural Science

March 2020 – May 2024

Sugar Land, TX

- Independently led and managed summer camp classes in the instructor's absence, demonstrating leadership and adaptability
- Developed lesson plans and educational materials for science-focused programs
- Mentored students and junior staff, fostering an engaging learning environment

Projects

Tensor Compiler Optimization Pipeline

Compiler Construction Project, University of Oklahoma

Fall 2025

Norman, OK

- Built a schedule-driven optimization pipeline that transforms YAML-defined tensor kernels through AST lowering passes into executable C
- Implemented hoisting, modulo simplification, pointer-dereference lowering, and kernel-to-SSA passes to improve generated code quality
- Created CLI verification and benchmarking workflows to validate correctness and compare optimization variants reproducibly

Classroom Q&A

Personal/Academic Project

March 2025 – May 2025

Norman, OK

- Developed and deployed a full-stack anonymous classroom Q&A app used in a live university course environment
- Implemented real-time sync with Firebase/Firestore, TypeScript, React, Next.js, and Tailwind CSS to boost student participation

FreshStart (2nd Place, Hacklahoma 2024)

Hacklahoma

Feb. 2024

Norman, OK

- Achieved 2nd place out of 200+ participants while leading front-end development for an award-winning farm planning web app
- Built interactive farm planning UI with MongoDB, Express, React, and Node.js (MERN stack)
- Integrated machine learning for crop yield prediction and an LLM for personalized farm task generation

Leadership & Activities

Pride of Oklahoma Marching Band Member

University of Oklahoma

Aug. 2023 – Dec. 2024

Norman, OK

- Balanced a rigorous rehearsal and performance schedule with full-time coursework, collaborating under high-pressure game-day conditions